



IDX EXCHANGE

Single-Family Home Valuation Project

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Overview

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Objective

Develop a robust AVM to predict California single-family home sales prices from monthly CRMLS sales data, accounting for:

- Regional market differences
 - Coastal metros vs. inland valleys
- Data quality challenges
 - Invalid records
 - Missing sales information
 - Extreme outliers
- Generalizability
 - Applicable to **both on- and off-market** properties

Preprocessing

1

Data Validation

Applied to all available data

2

Data Transformation

Applied after the chronological split to prevent data leakage:
learn transformations from the training set and apply them
unchanged to validation and test sets

Data Validation*

Remove invalid sales records to ensure reliable model inputs

- Is the sales timeline logical?
 - Listing date $\leq$ purchase date $\leq$ close date
- Is the property located in California?
- Are the property characteristics plausible?
 - Beds, baths, stories, lot size, parking, garage spaces ≥ 0
 - Cross-check boolean indicators to interpret missing values correctly
- Are there duplicate records?

*This step also removes features that approximate close price, reflect pricing strategy, contain post-close information, or introduce target leakage in other ways

Data Transformation

Denoise, impute, encode, scale, and enrich existing data to make it more suitable for modeling

- Remove price outliers
 - Close price → close price per sq ft
- Perform groupwise imputations
 - Living area, lot size, property age by county
 - Baths, stories, parking space by living area
- Log-transform close price
- Encode categorical features
 - One-hot encode boolean indicators
 - Target encode high-cardinality location features
- Derive new features
 - Property age from built year
 - School district from geographical coordinates

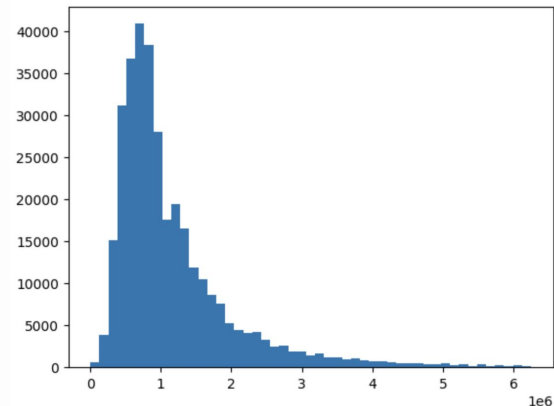
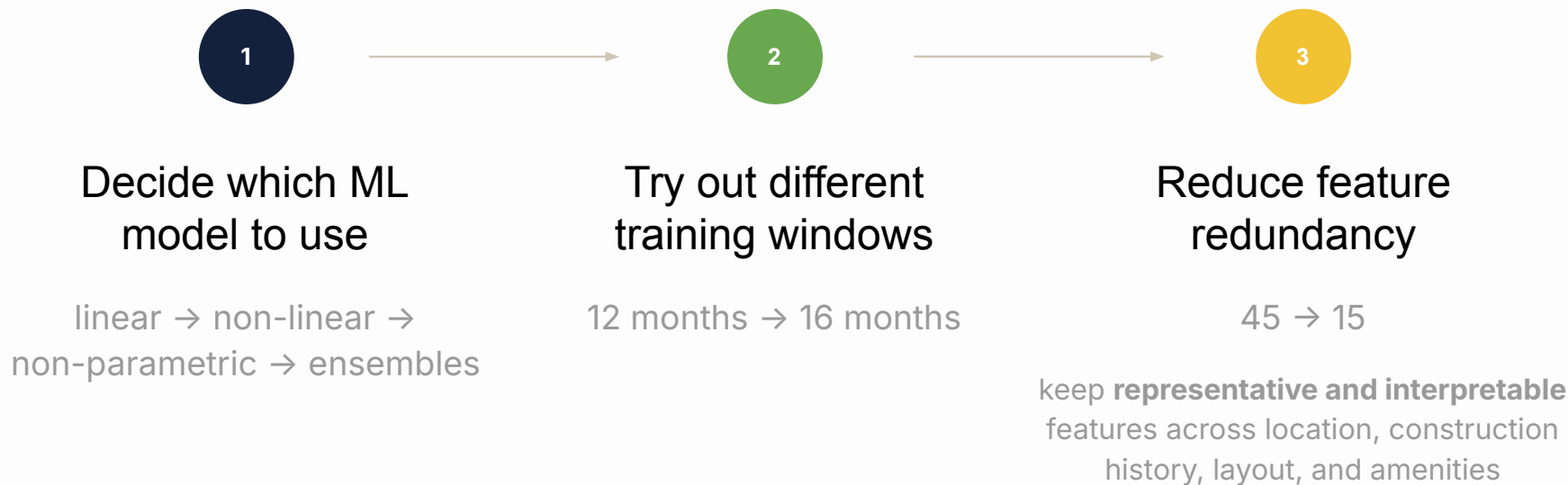


Figure 1: Distribution of close prices, January 2024–June 2026, showing notable right skewness

Iteration Timeline

Model tuning* proceeds in three stages:



*Models are tuned to minimize validation MdAPE

Performance Comparison

Model	MdAPE	MAPE	R ²
XGBoost	7.66%	11.41%	0.89
LightGBM	7.85%	11.24%	0.90
Random Forest	8.36%	12.02%	0.88
Natural Spline	11.15%	15.13%	0.84
Linear Regression	12.38%	16.36%	0.80

Figure 2: Top-performing model from each category based on **validation MdAPE**; XGBoost and LightGBM stand out with MdAPE below 8%

XGBoost and LightGBM achieve:

- Consistent predictive accuracy on the test set
 - MdAPE declines at higher price quartiles
- Stable performance across rolling backtests
 - XGBoost: MdAPE SD = 0.16%, MAPE SD = 0.25%, R² SD = 0.0064
 - LightGBM: MdAPE SD = 0.11%, MAPE SD = 0.24%, R² SD = 0.0057

Takeaways

Domain-specific preprocessing is essential beyond standard data cleaning

Flexible non-parametric ensemble models generally perform better, though training window length and feature set also factor into predictive accuracy

*LightGBM is selected for deployment over XGBoost due to its superior computational efficiency and predictive stability. Try the app here:
<https://ca-single-family-home-value-estimator.streamlit.app/>

Thank You!

Got questions? Ask away!